Maximus Johnston

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Highly motivated Data Scientist with a proven record of delivering impactful analytics and machine learning solutions at scale across government, retail, and technology sectors. Skilled at uncovering insights in complex, large-scale datasets, shaping strategy, and securing stakeholder buy-in for transformative initiatives. Recently graduated cum laude with a master's in data science from the University of Amsterdam, advancing technical expertise and applying cutting-edge methods to real-world challenges. I am now looking for new exciting challenges.

Work Experience

Data Scientist (April 2023 – August 2025) - ***Department for Work and Pensions (DWP)***

Forecasting & Predictive Modelling *(PySpark, scikit-learn, Pandas, git)*

- Built and deployed forecasting models with >90% accuracy in predicting health service uptake, informing strategic workforce planning for thousands of staff and optimising multi-million-pound resource allocation

Advanced Analytics & Insight Discovery *(scikit-learn, NLP with Python, Pandas, git)*

- Discovered hidden claimant health-benefit patterns using K-means clustering and unstructured text analysis, enabling targeted interventions that improved service delivery for tens of thousands of claimants.

Stakeholder Engagement & Communication *(PowerBI, Looker, Matplotlib, git)*

- Partnered with operational and software teams to translate business needs into actionable data products, increasing adoption of data-driven decision-making across multiple programmes.
- Designed clear, accessible visualisations that accelerated programme-level decisions, reducing analysis-to-action time from weeks to days.

Transforming Audit and Operational Efficiency *(SQL, Python, Databricks, git)*

- Scoped a high-value audit improvement project for health assessments, identifying opportunities for substantial savings for taxpayers and uncovering a reusable BAU process with a potential 6 figure saving
- Collaborated across audit, operational, and policy teams to secure senior approval for implementation, influencing national process changes.

Data Science Intern (May - July 2021) - ***Data Clarity Limited, for a leading global British Fashion company***

Insight Generation from Large-Scale Retail Data *(Python, Pandas, Matplotlib, SQL, NetworkX, git)*

- Analysed 3M+ transactions to generate actionable insights, improving product recommendations and increasing targeted marketing effectiveness.

Predictive Modelling for Customer Behaviour *(scikit-learn, Pandas, Xgboost, SQL, git)*

- Developed predictive models to estimate next purchase dates with high accuracy, driving measurable sales uplift through optimised advertising and product placement

Undergraduate Software Engineer (September 2020- May 2021) - ***XACTPCB LTD,*** *(C#, XAML, WPF, git)*
Data Science Research Intern (July 2020- August 2020) - ***ENGIE*** *(Python data stack)*
Software and solutions architecture Intern (June - August 2019) - ***Excel Redstone/Google*** *(Java)*

Selected Projects

MSc Thesis: Demand Estimation under Stockouts in Fashion Retail (Python, Polars, PyTorch) - ***University of Amsterdam & WAIR AI***

- Benchmarked and extended the EM-MNL demand estimation model on sales data from Dutch fashion stores with stockout censoring, showing through Monte Carlo simulation that EM does not converge to the exact maximum likelihood estimate under heavy censoring, and developed DiffMNL, a neurosymbolic extension that models substitution between products.

Multi-Agent LLM Marketing Platform - ***Personal project***

- Designed a multi-agent generative AI platform for marketing, supported by a business plan reviewed as part of the course.

NLP Projects - ***University of Amsterdam***

- Received highest grade in cohort for applied NLP methods to UN General Debate voting data and built a BiLSTM classifier for logical reasoning.

Skills

Python, Python Data Stack, PySpark, SQL, Git, C#, Looker, Power BI, Java, Databricks, scikit-learn, XGBoost, Matplotlib, NetworkX

Education

Universiteit Van Amsterdam (August 2025 – August 2026)

- Master's in Data Science (Information Studies), graduated cum laude (with distinction).
- Built expertise in statistical modelling, AI, and large-scale data engineering.

Northumbria University (2023 - 2025)

- Postgraduate Diploma in Digital and Technology Solutions (Data Analytics pathway) – master's-level apprenticeship combining advanced data science with the creation and communication of innovative technological strategies and business proposals to senior stakeholders

Newcastle University (2018 - 2022)

- First-class degree in Computer Science with industrial placement (BSc). Including a first-class dissertation on the application of federated learning models created using TensorFlow

Portfolio: https://maximusj08.github.io/